

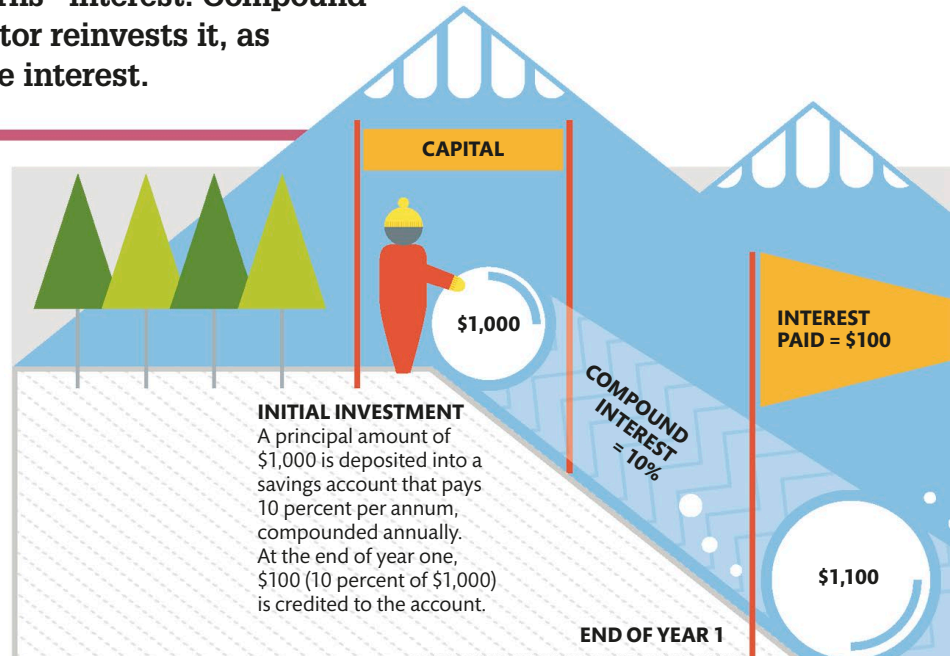


Interest and compound interest

When money is saved it “earns” interest. Compound interest accrues if the investor reinvests it, as opposed to withdrawing the interest.

The snowball effect

If a snowball is rolled down a hill, it gets bigger and bigger as it gathers more snow. In addition, the rate at which the snowball grows increases as it rolls down the hill because there is a greater surface area for the snow to stick to. So, given enough time, a tiny snowball can become a giant one. Compound interest has been described as a “snowball effect” as it works in much the same way, meaning that a small investment can provide bigger returns than an investment where the interest sum is paid out to the investor annually.



INITIAL INVESTMENT

A principal amount of \$1,000 is deposited into a savings account that pays 10 percent per annum, compounded annually. At the end of year one, \$100 (10 percent of \$1,000) is credited to the account.

END OF YEAR 1

INVESTMENT GROWS

The savings account now has \$1,100, then earns \$110 (10 percent of \$1,100) interest in the second year. By the end of year two the account has a balance of \$1,210.



NEED TO KNOW

- **Principal amount** The original capital sum invested or borrowed.
- **Compounding frequency** The number of times that interest is added to the principal amount in one year. For example, if interest is added monthly, the compounding frequency is 12.
- **Annual percentage rate (APR)** The annual rate of interest payable on mortgages, credit cards, etc. It includes the rate as well as fees, such as annual fees. APR can be referred to as the annual equivalent rate (AER), effective interest rate (EIR), or effective annual rate (EAR).



Compound interest formula

“A” is the final amount in the savings account after “T” years’ interest compounded “N” times, at interest rate “R” on the starting amount of “P”.

$$A = P \left(1 + \frac{R}{N} \right)^{NT}$$

Final amount

Rate of interest

Time in years

Principal (original) sum

Number of times per year interest is compounded